



JAI AUTO PVT. LIMITED

VILLAGE KADDON. G.T ROAD DORAHA LUDHIANA (P.B.)INDIA

E-Mail:info@jaiauto.in

Works Test Certificate

EN 10204/3.1

JAPL/LAB NO: 23-24/

INVOICE NO.: (J2324-114)

DATE: 05-09-2023

Purchase Order No 354247

Customer

DODSON GLOBAL
2750 HOLMES ROAD
HOUSTON, TX 77051
U.S.A.

Forging specification

ASME SA 105-2019
ASTM A105-2021

Component 4" -900# BLRF FLG.

Item Code

Steel Supplier

JAI JAWALA

Machining STD ANSI B16.5- 2020
ASME B16.5- 2020

Quantity

63Nos.

Mill Heat No.

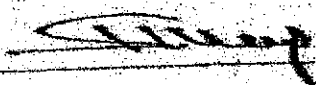
AUG=21

TEST RESULTS

JAPL. HEAT NO: 2502

CHEMICAL COMPOSITION

MECHANICAL PROPERTIES

Element %	Specified Composition	Actual		PROPERTIES	SPECIFIED	OBTAINED	Specified carbon as per Purchase Order 0.25max.
		JAPL LAB	MILL T.C.				
Carbon	0.35max	0.19	0.20	Yield Strength (mpa)	250.00min.	341.62	Serrated face as per MSS-SP-6 With in limits of 125-250AARH
Manganese	0.60-1.35	1.02	1.03				
Silicon	0.10-0.35	0.31	0.34				
Sulphur	0.040max	0.004	0.016	Tensile Strength (mpa)	485.00min.	598..68	H.T.:- Normalized at temp. 920°C
Phosphorous	0.035max	0.024	0.027				
Chromium	0.30max	0.031	0.039	%Elongation in 50mm / 4D	22.00min.	34.10	CARBON EQUIVALENT (CE) CE = C + $\frac{Mn}{6}$ + $\frac{Cr}{5}$ + $\frac{Mo}{5}$ + $\frac{V}{15}$ + $\frac{Ni}{15}$ + $\frac{Cu}{15}$
Nickel	0.40max	0.013	0.06	Hardness HBW	187max.	161-167	
Copper	0.40max	0.019	0.032				
Molybdenum	0.12max	0.010	0.020	% Reduction of Area	30.00min.	63.71	MARKING AS PER MSS-SP-25-2013
Vanadium	0.08max	0.002	0.01	Remarks:- We hereby certify that the product satisfies the requirement of ASTM A 105 and material described here in has been tested in accordance with the applicable specifications referred above and meets the requirements as per EN10204/3.1B & NACE MRO-175 & ISO 15156. Product does not come in contact with mercury and lead during the manufacturing process. No weld repair was performed and all products are free of weld repair			4-900 SA/A105 N2502 B16.5 JAPL INDIA
Carbon Equivalent (CE)	0.47max	0.371	0.392				
							Checked by: 
							Metallurgist
							Certified by: 

Checked by:

Metallurgist

Certified by:

Quality Control Manager

4" RF Blind Flange